

ELEC3130 Spring 2026

Digital Image Processing

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The Hong Kong University of Science and Technology
Department of Electronic and Computer Engineering

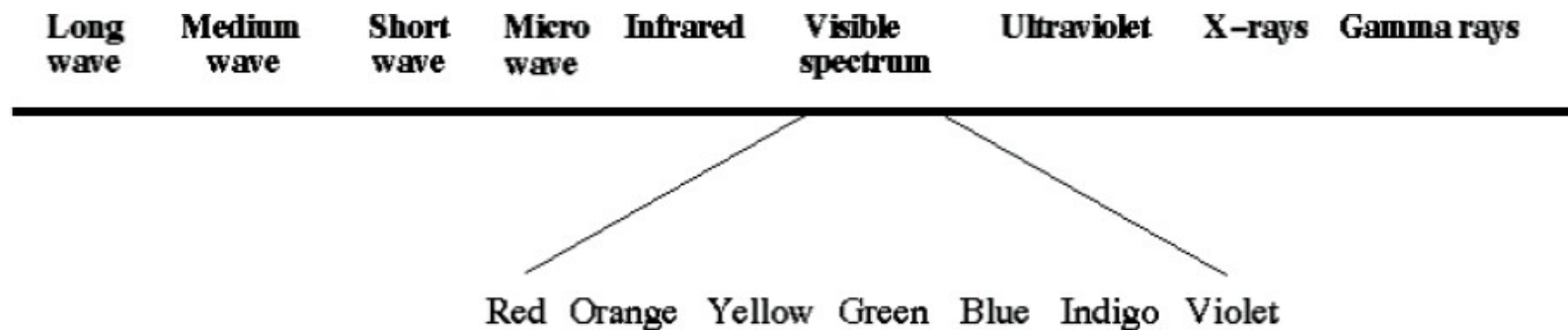
- ▶ Class hours:
Wed/Fri 4:30PM-5:50 PM
 - ▶ Room: 5506
 - ▶ Office hour: Wed 6:00-6:30PM
Office: 2444
 - ▶ Email: eewangyw@ust.hk
- ▶ Teaching Assistant:
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zwangfr@connect.ust.hk
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 - ▶ TA Hour: Wed
1:00pm-2:00pm
 - ▶ Room: 4218D

Overview

- ▶ What is Digital Image?
- ▶ What is Processing on Images?
- ▶ Why is it important?
- ▶ What knowledge do you need in order to take this course?
- ▶ What is going to be covered in this course?
- ▶ What textbooks are we going to use?
- ▶ Where to obtain additional information?
- ▶ How are we going to teach this course?

What are Images?

- ▶ (Sets of) Objects perceived by Human Visual System (HVS)
- ▶ Light waves Emitted or Reflected from Objects are Received and Processed by Eye and Brain.
- ▶ Light waves are Electro Magnetic waves



What is visual system?

Vision

Visual information dominates perceptions and frame the way we think.

Vision pathway

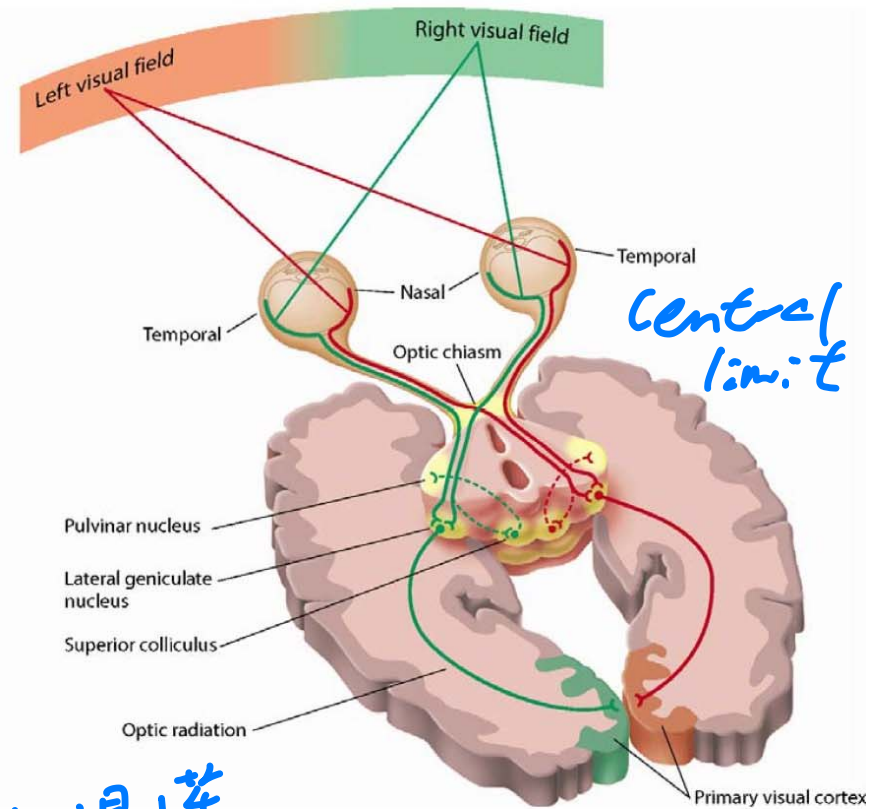
- Photoreceptors
 - Retina: cones (color) and rods (night vision)
- Optic nerve
 - Ganglion cells, compress the information
- Lateral geniculate nucleus (LGN)
- Primary visual cortex (V1)

2 eyes $\Rightarrow$ see depth!

see $\neq$ understand

不一定看得懂

$\Rightarrow$ recognize!



What are Images?



- ▶ Image: a visual representation in form of a function $f(x, y)$ where f is related to the brightness (or color) at point (x, y)
- ▶ Most images are defined over a rectangle
- ▶ Continuous in amplitude and space

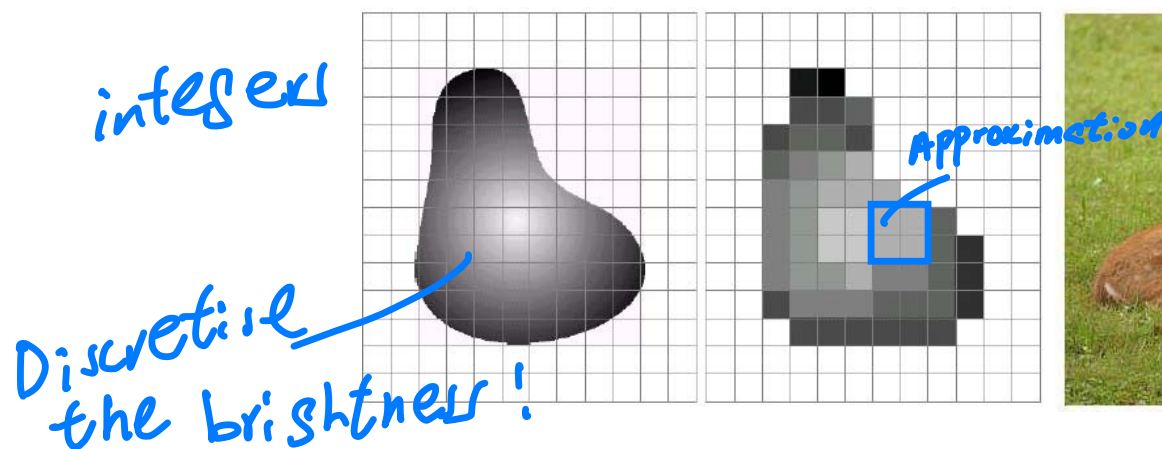
brain no pixel

2D $\rightarrow$ pixel

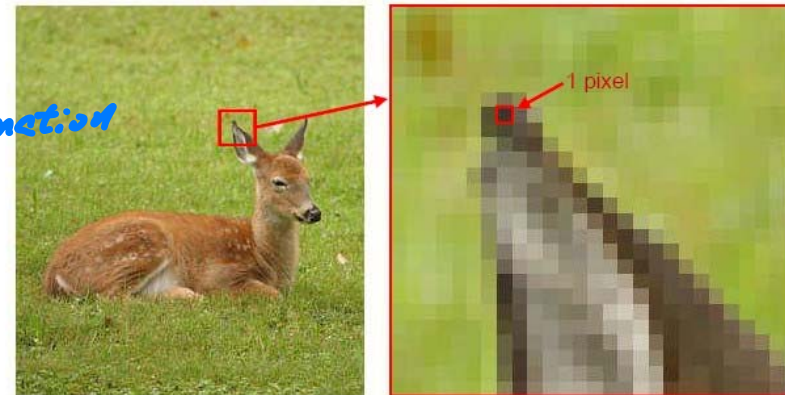
What is Digital Image?

digital $\Rightarrow$ easy to save, restore!

- ▶ Discrete samples $f[x, y]$ representing continuous image $f(x, y)$. f , x , and y are all finite, discrete quantities
- ▶ Each element of the 2-d array $f[x, y]$ is called a pixel
- ▶ Remember digitization implies that a digital image is an approximation of a real scene



Artificial!

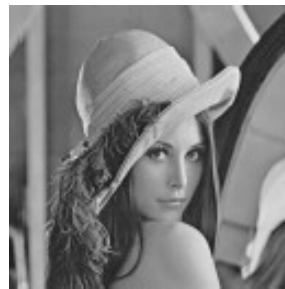
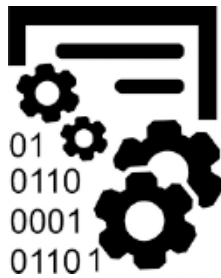


- ▶ Knowledge in ELEC2100 (signals and systems)/3100 (signal processing and communications) will be helpful

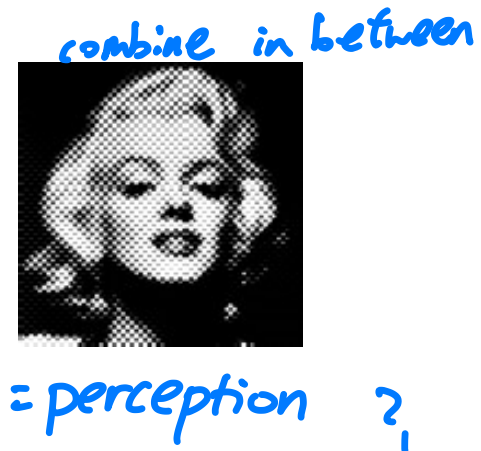
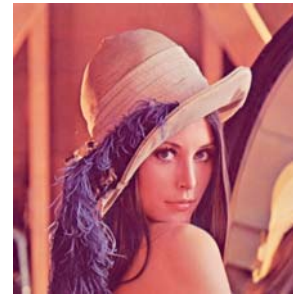
Spatial transform *signal in space!*

Basic Types of Digital Images

- ▶ Binary: (Black and White) Only two kinds of pixels
- ▶ Gray Scale: Several levels of gray (Ex. 8 bit gray scale has 256 different gray levels)
- ▶ Color: Usually 3 components
- ▶ Half-tone: Gives effect of gray scale using only black and white pixels



easy
←

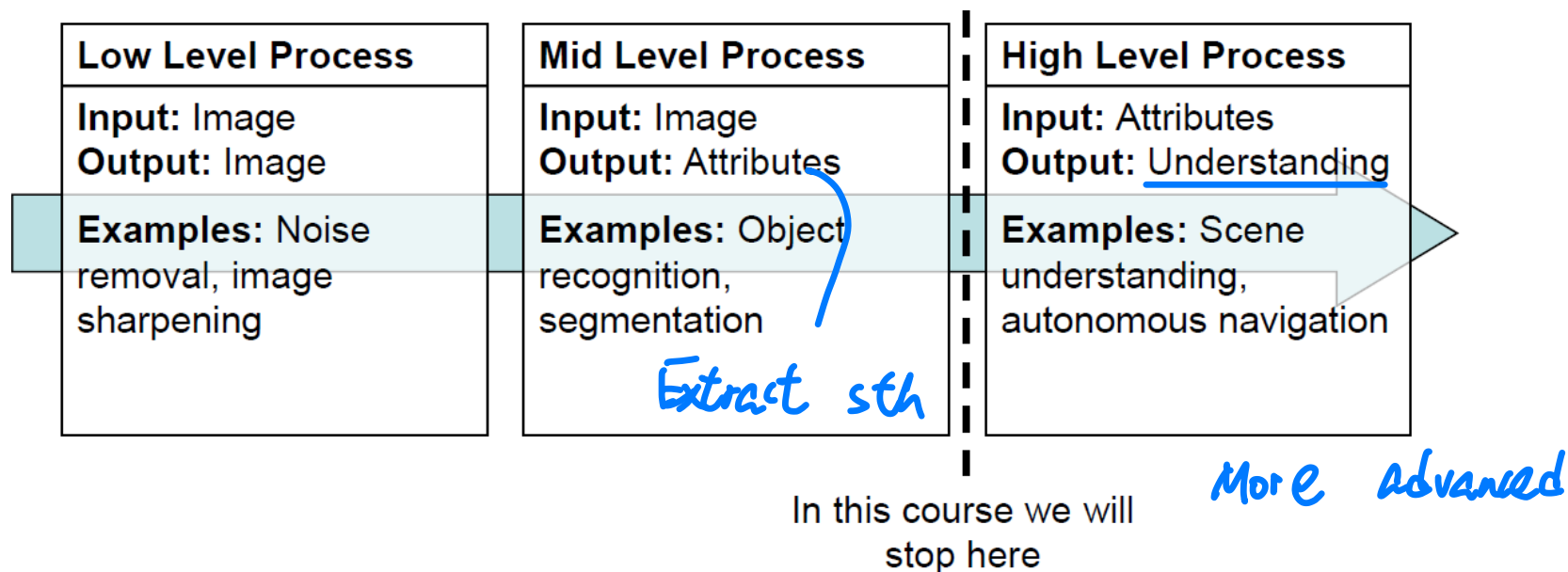


What is Processing on Digital Image ?

- ▶ Focuses on two major tasks
 - ▶ Improvement of pictorial information for human interpretation *subjective!*
 - ▶ Processing of image data for storage, transmission and extraction of pictorial information *buy more :)*
- ▶ Some argument about where image processing ends and other fields, such as image analysis and computer vision, start

What is Processing on Digital Image? (cont.)

- ▶ The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes



Importance of Digital Image

- ▶ Visual perception is the most important source of information for human being
 - ▶ Seeing it once is more than hearing it hundred times
 - ▶ I see (not I hear nor I touch)
- ▶ Spatial structure information in the perception
- ▶ Digital image has shown many important applications

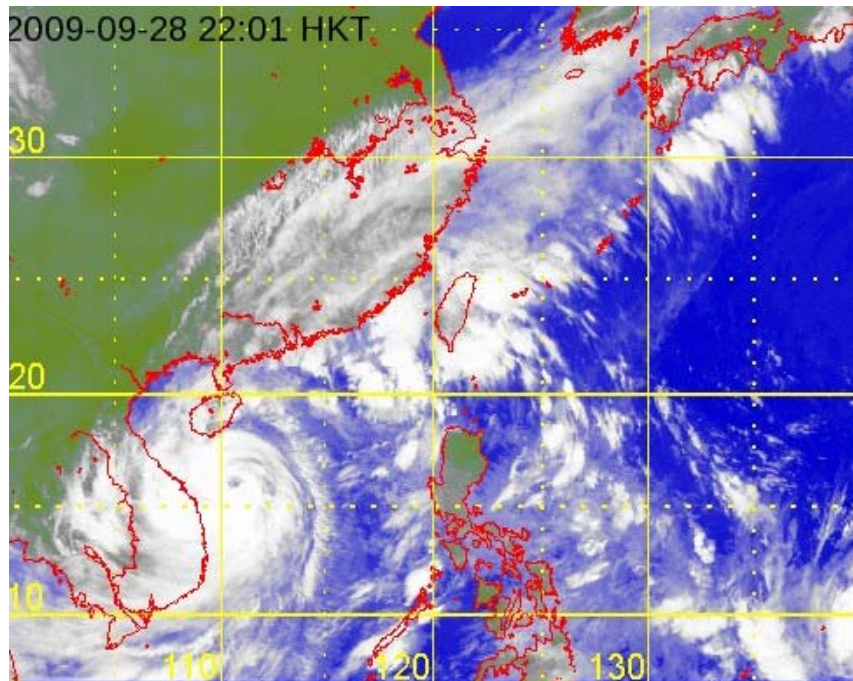
Why processing digital images?

- ▶ Acquire an image!
 - ▶ Correct aperture and color balance
 - ▶ Reconstruct image from projections
- ▶ Prepare for display or printing!
 - ▶ Adjust image size
 - ▶ Color mapping, gamma-correction, halftoning
- ▶ Facilitate picture storage and transmission!
 - ▶ Efficiently store an image in a digital camera
 - ▶ Send an image from space
- ▶ Enhance and restore images!
 - ▶ Touch up personal photos
 - ▶ Color enhancement for security screening
- ▶ Extract information from images!
 - ▶ Read 2-d bar codes
 - ▶ Character recognition



Examples of Images

- ▶ Remote sensing and weather forecast

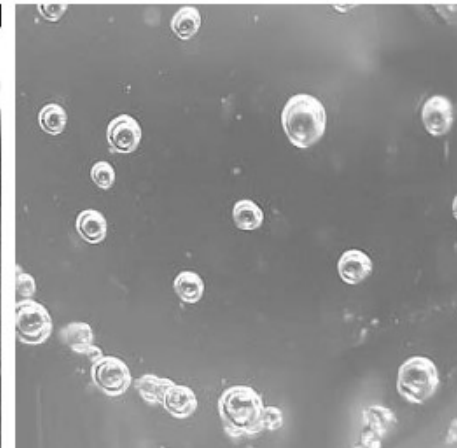
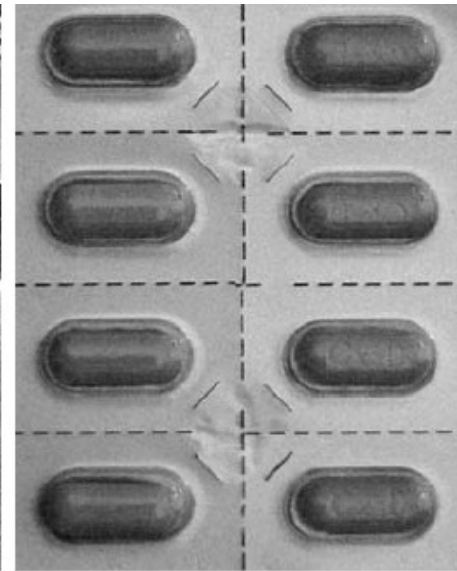
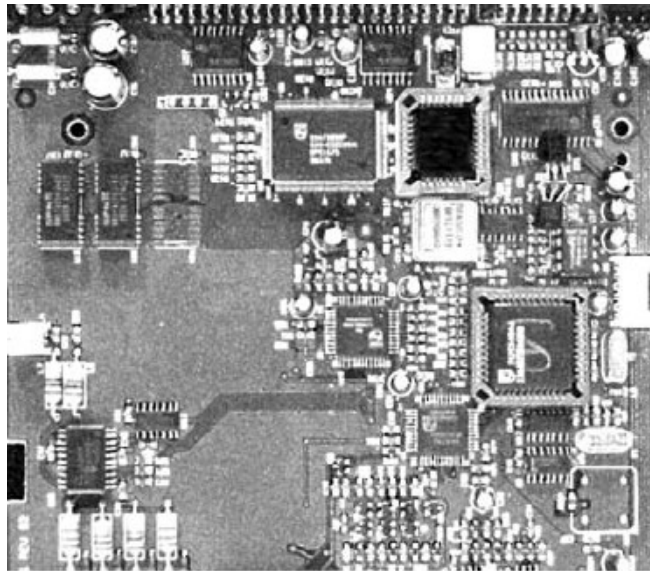


- ▶ Law Enforcement



Examples: Visual inspection of manufactured goods

*check
circuit!*

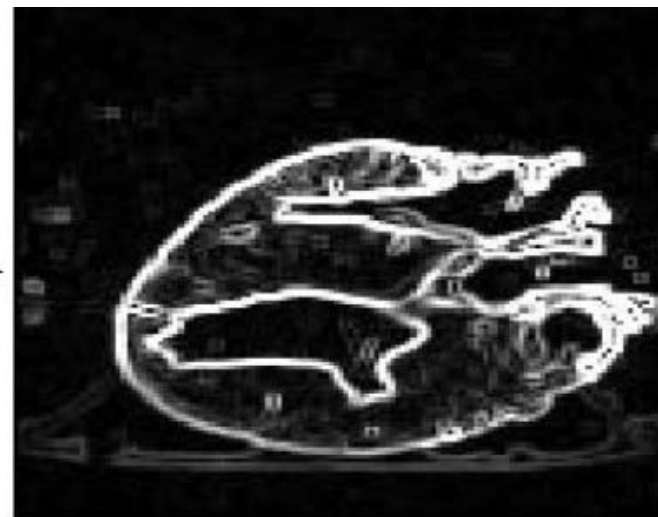


Examples: Medical Imaging

- ▶ Take slice from MRI scan (in radio band) of canine heart, and find boundaries between types of tissue
 - ▶ Image with gray levels representing tissue density
 - ▶ Use a suitable filter to highlight edges



Original MRI Image of a Dog Heart



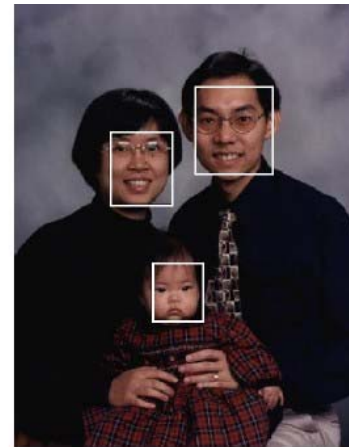
Edge Detection Image

Examples: Artistic Effects

- ▶ Artistic effects are used to make images more visually appealing, to add special effects and to make composite images



Examples: Face detection



source: Henry Chang, Ulises Robles, EE368 class project, spring 2000.

Examples: Face Blurring



source: Henry Chang, Unises Kovies, EE500 class project, spring 2000.

Examples: Paint recognition



source: Henry Chang, Onses Kobles, EE506 class project, spring 2000.

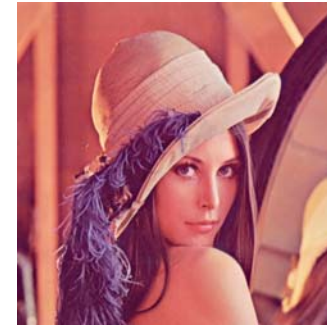
Examples: CD recognition



Image $\neq$ Information *(only human interpretation)*

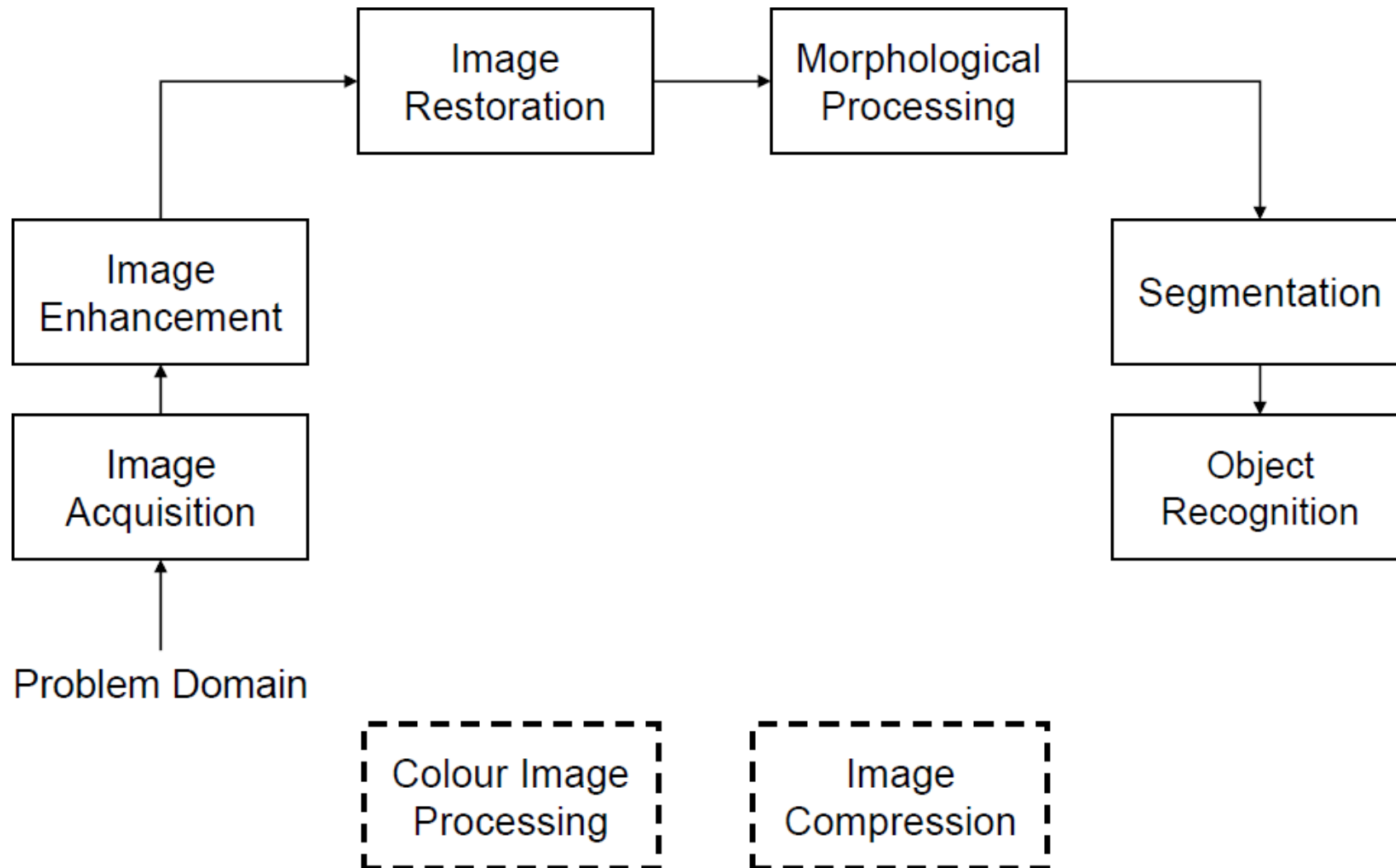


resizes
→

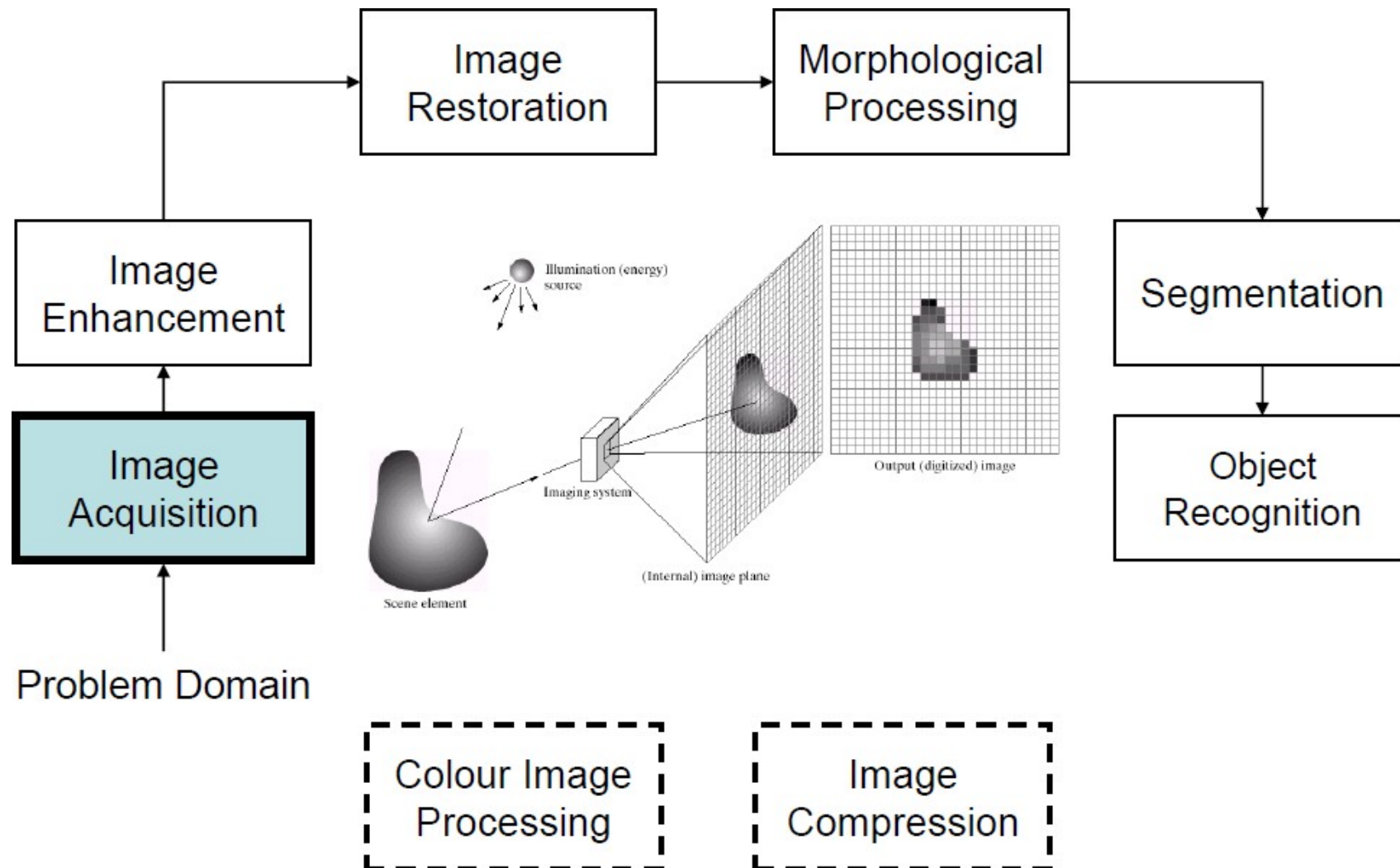


- ▶ Same information in the above two images?
- ▶ How to measure the information quantitatively?
- ▶ What processing do we need to extract information from images?
- ▶ Interesting story at <http://www.lenna.org/>

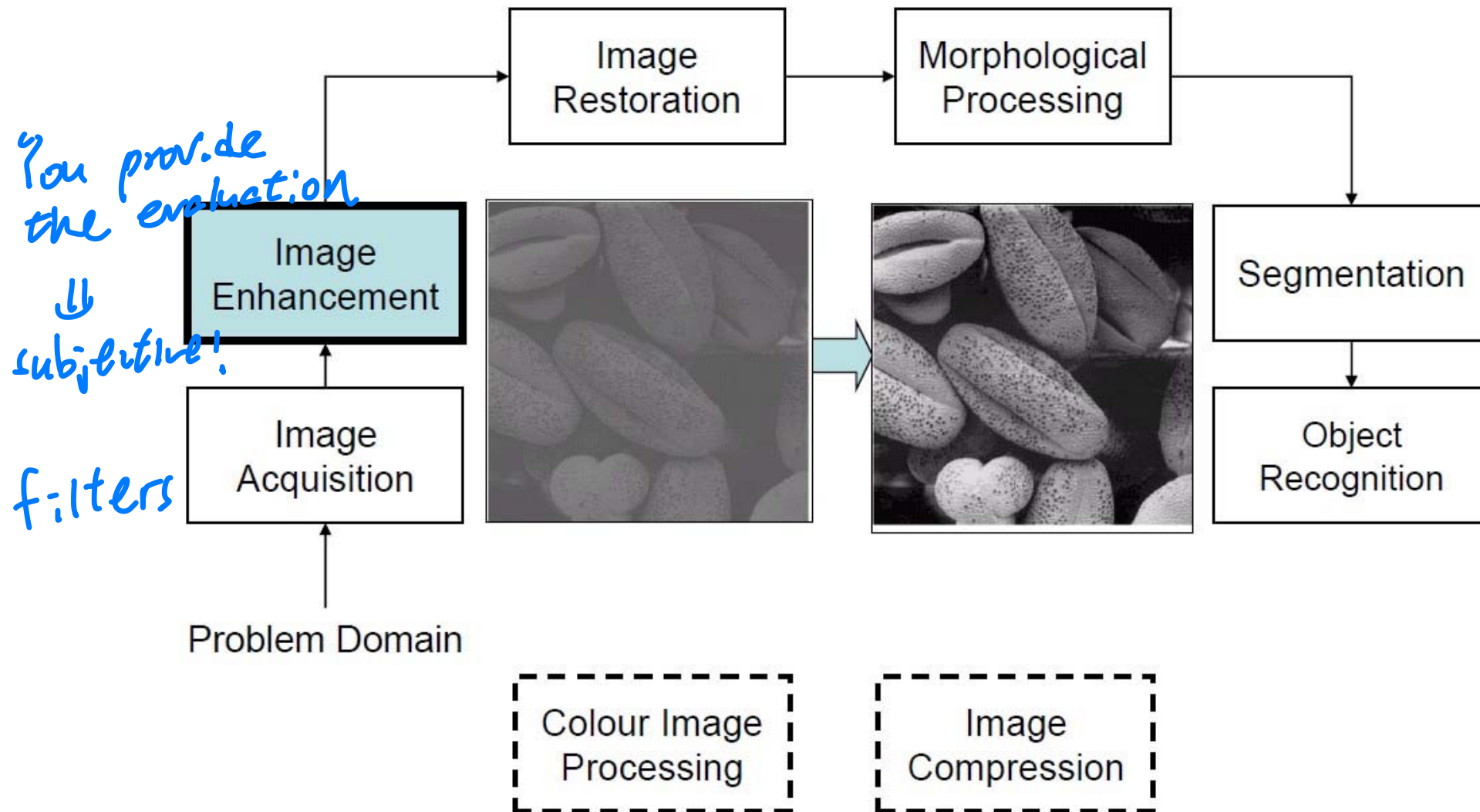
Key stages



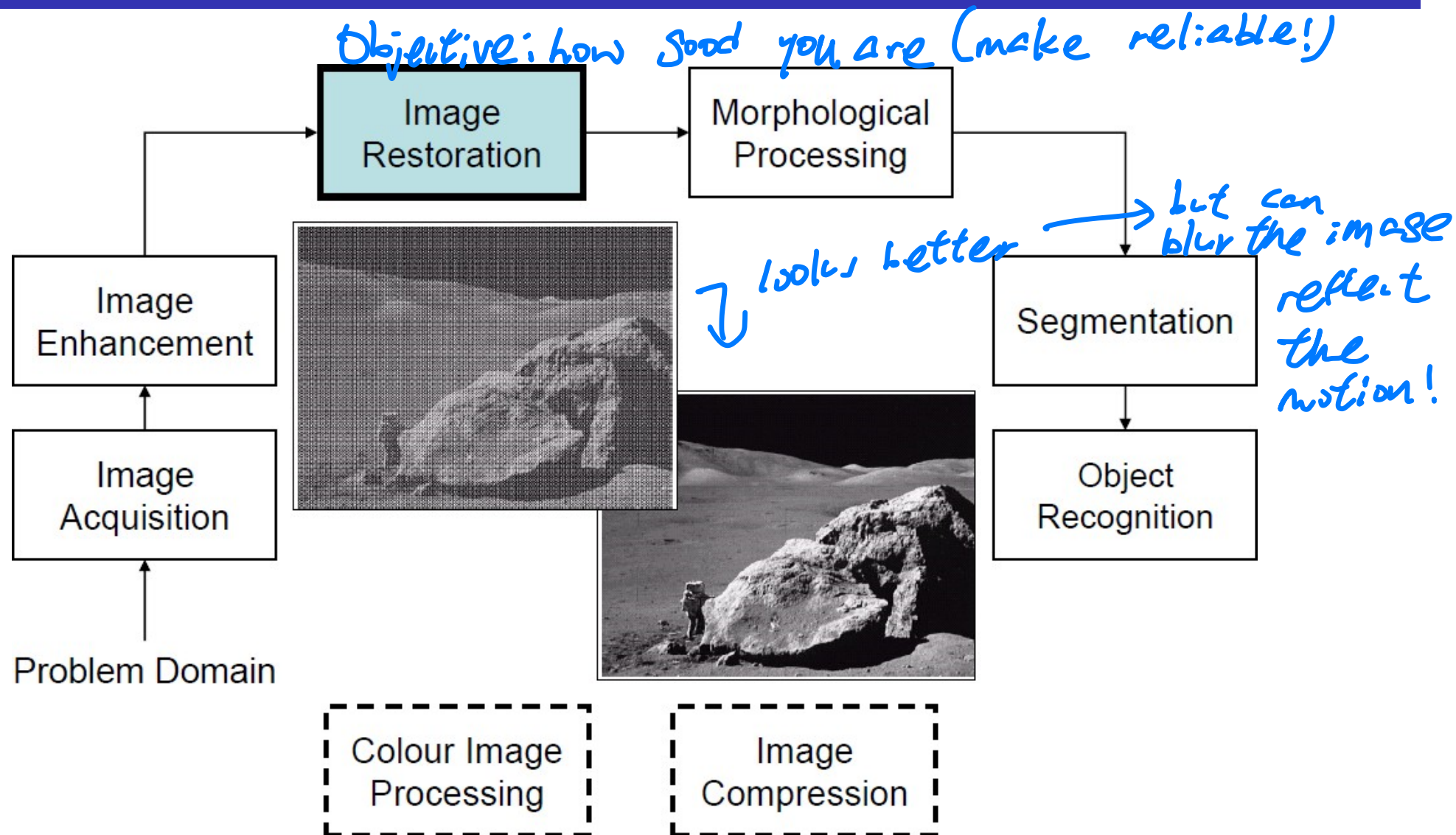
Key stages: Image Acquisition



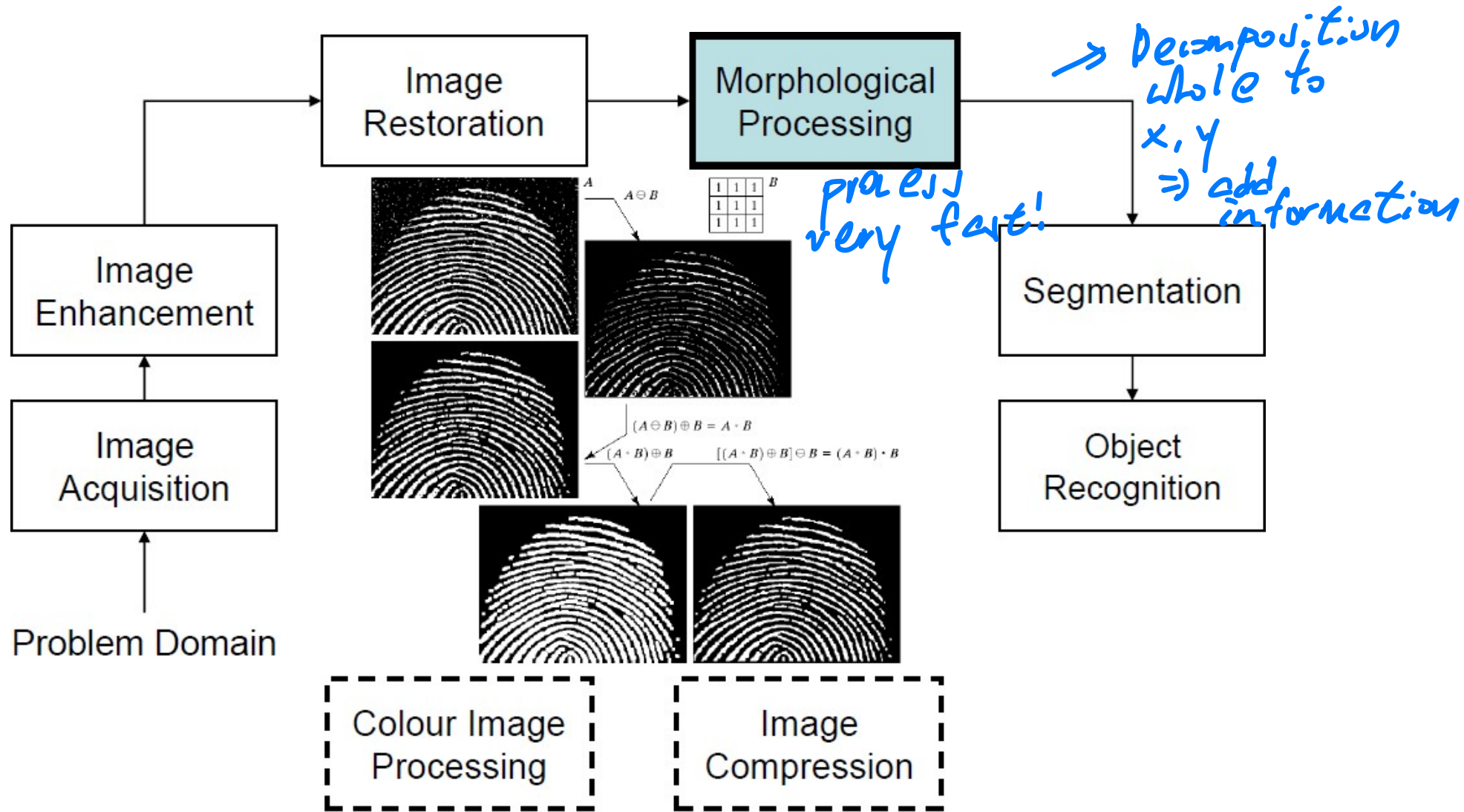
Key stages: Image Enhancement



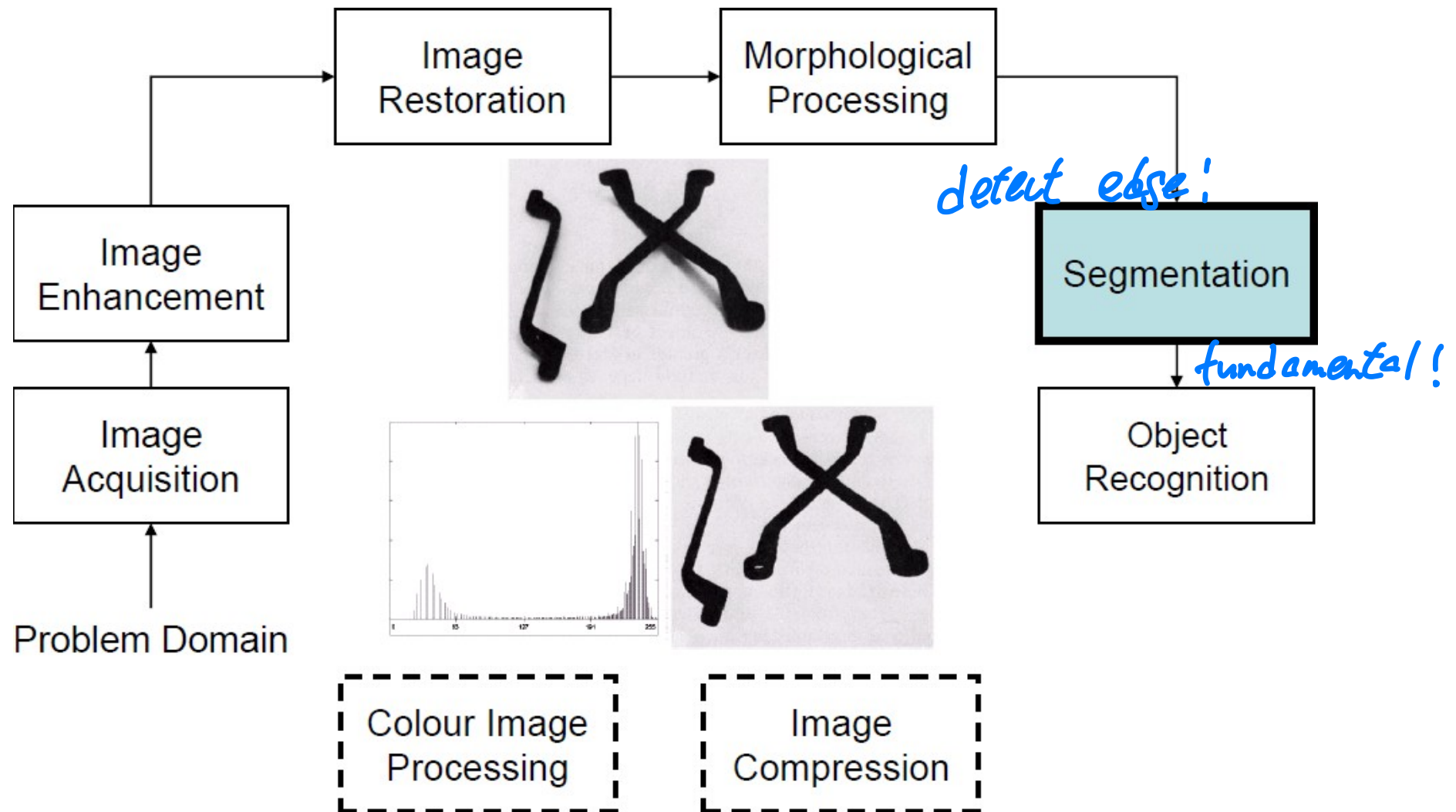
Key stages: Image Restoration



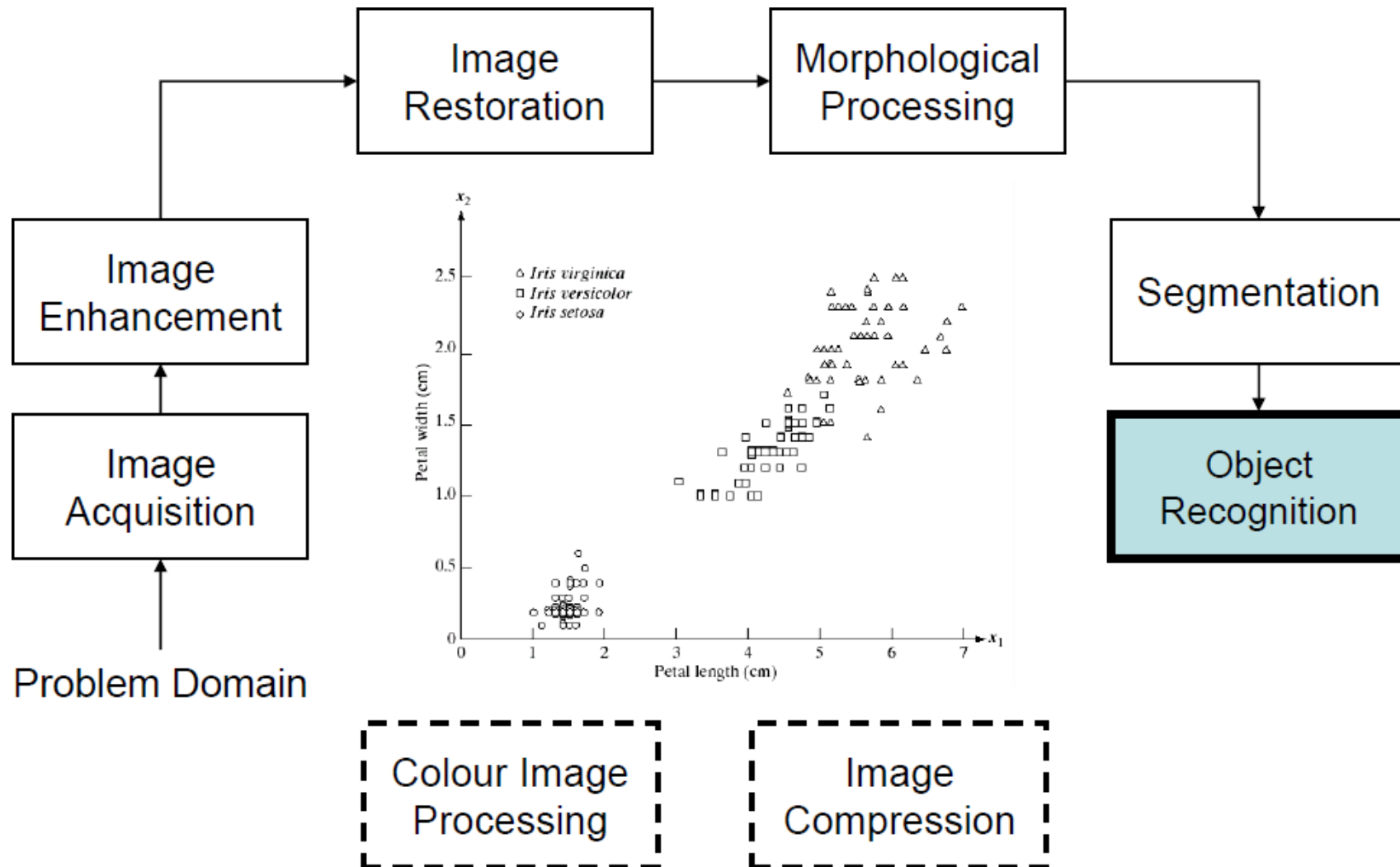
Key stages: Morphological Processing



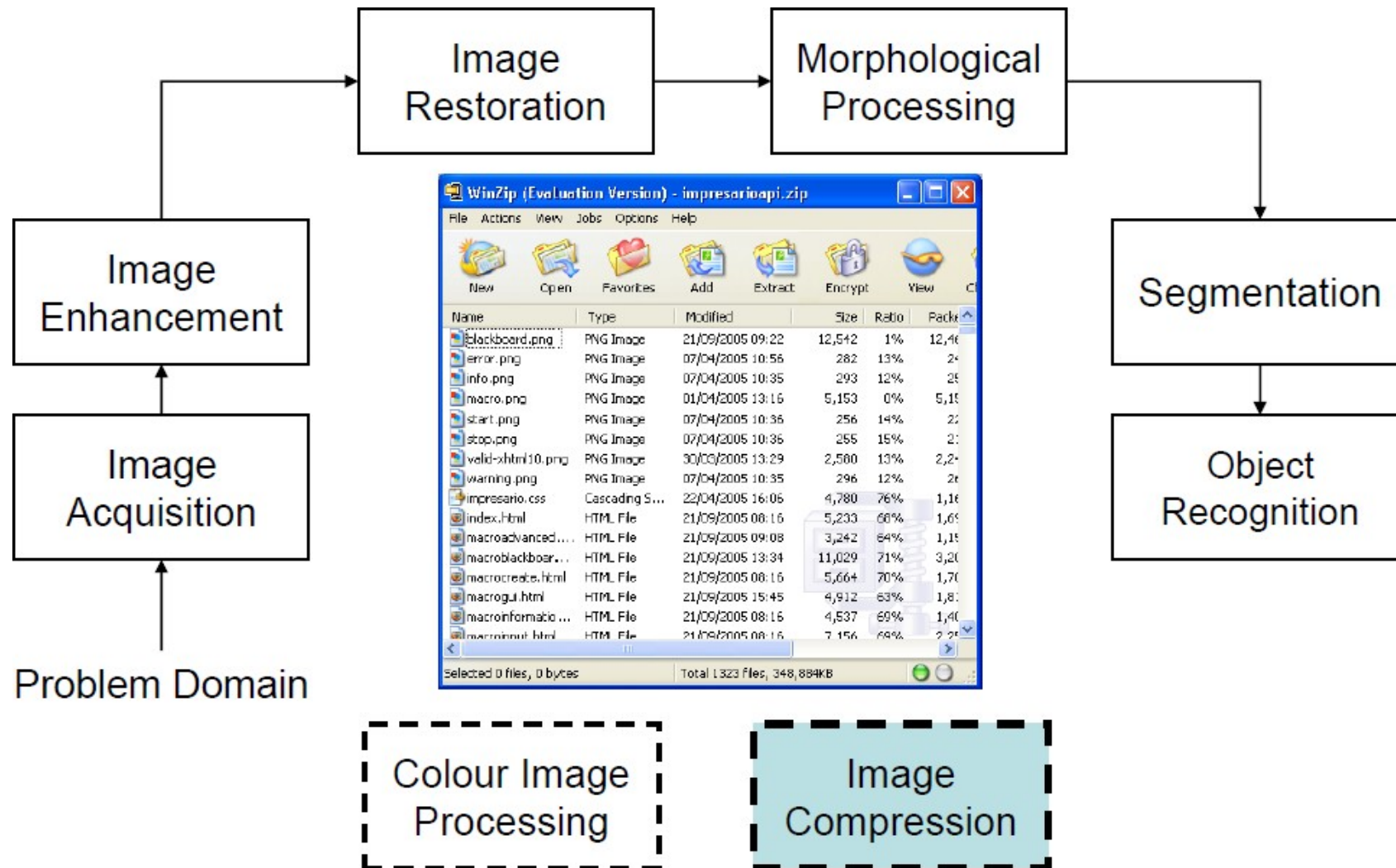
Key stages: Segmentation



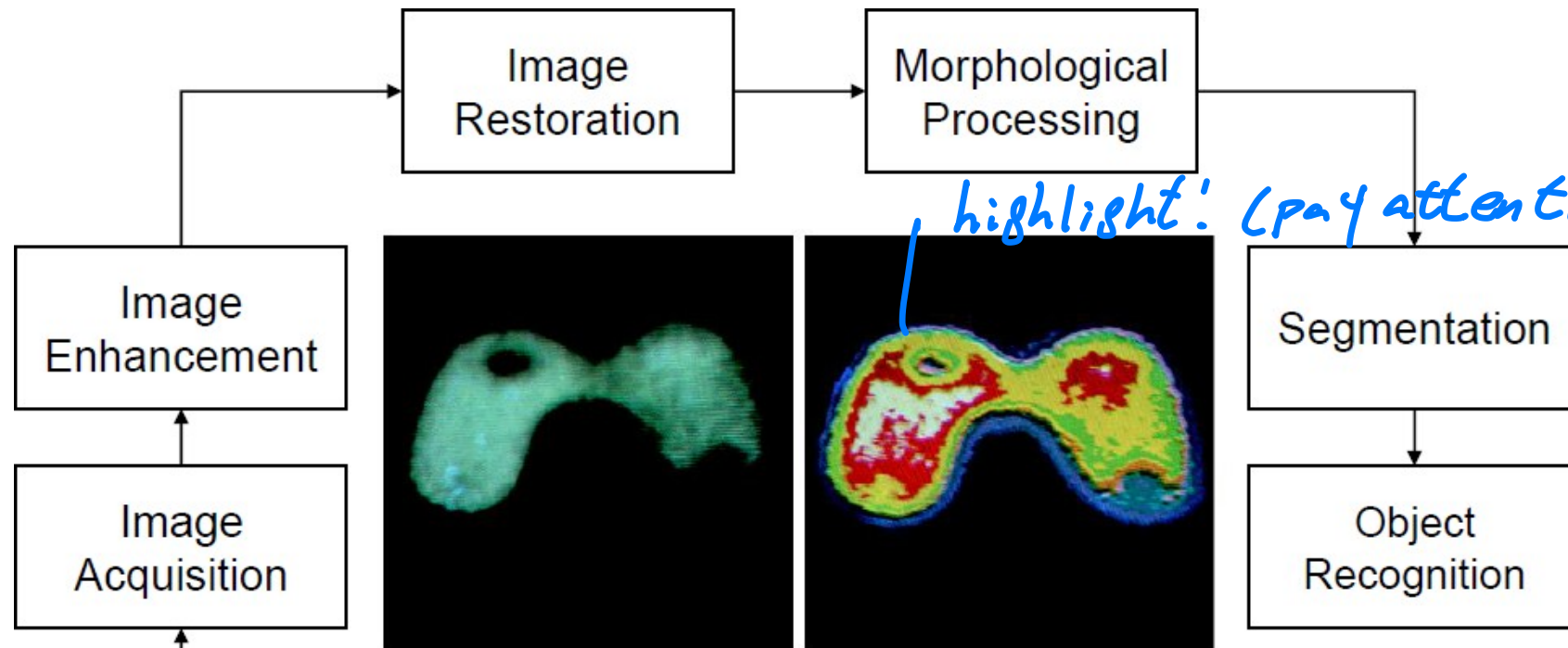
Key stages: Object Recognition



Key stages: Image Compression



Key stages: Colour Image Processing



Decompose
the colour!

Colour Image
Processing

non-linear
transform

Image
Compression

Course Arrangement

Week 1	Introduction
Week 2	Transforms
Week 3	Morphological Image Processing
Week 4	Image Enhancement in the Spatial Domain
Week 5	Image Enhancement in the Frequency Domain
Week 6	Image Restoration
Week 7	Motion Deblurring using Wiener Filter
Week 8	Color Image Processing — Mid-term Exam (T.B.A.)
Week 9	Image Compression: JPEG as an Example
Week 10	Image Segmentation: Canny Edge Detector
Week 11	Object Recognition: Classical Approaches
Week 12	Face Recognition: Eigen Faces
Week 13	Final Review and Outlook: Deep Learning

► No class on Apr. 2, Apr. 4, Apr. 18

Prerequisite Knowledge for This Course

- ▶ Mathematics
 - ▶ Linear Algebra: Vectors and matrices
 - ▶ Probabilities: basic distributions, random variables
 - ▶ Signals and Systems: Transforms, linear systems, filtering, sampling (ELEC 2100)
 - ▶ Digital Signal Processing: ELEC 3100 or ELEC 3200
- ▶ Programming
 - ▶ MATLAB programming experience
- ▶ **This is a mathematics-oriented course!**
- ▶ If you are not familiar with above knowledge:

http://www.imageprocessingplace.com/root_files_V3/tutorials.htm

Teaching Style

- ▶ Textbook is the basic reading material
 - ▶ Digital Image Processing by Gonzalez and Woods , 4th Edition, 2017
<https://www.amazon.com/Digital-Image-Processing-Rafael-Gonzalez/dp/0133356728>
 - ▶ textbooks are reserved in the library
- ▶ Lecture notes cover key concepts
- ▶ Details will be explained on the WHITE BOARD
 - ▶ Your attendance is necessary (although not required) for you to learn the details.
 - ▶ Some details are provided in the additional reading material.
- ▶ Agreement of classroom behavior

Course Grading

- ▶ Homework: 20%
- ▶ Mid-term exam: 30%
- ▶ Final project: 50%
- ▶ No-cheating policy:
 - ▶ Do not copy other people's homework
 - ▶ Do not let other people copy your homework
 - ▶ Both parties will be penalized if found cheating

Additional Information

- ▶ Course webpage:

<https://canvas.ust.hk/courses/68288>

- ▶ Textbooks and tutorials:

<http://www.imageprocessingplace.com/>

- ▶ Software-centric books

- ▶ Gonzalez, Woods, Eddins, Digital Image Processing using Matlab, 2nd edition, Pearson-Prentice-Hall, 2009.
- ▶ Bradski, Kaehler, Learning OpenCV, OReilly Media, 2008.

- ▶ Comprehensive state-of-the-art

- ▶ AI Bovik (ed.), The Essential Guide to Image Processing, Academic Press, 2009.

- ▶ Journals/Conference Proceedings

- ▶ IEEE Transactions on Image Processing
- ▶ IEEE International Conference on Image Processing (ICIP)
- ▶ IEEE Computer Vision and Pattern Recognition (CVPR)

- ▶ MATLAB:

Mathworks webpage

www.mathworks.com/academia/student_center/tutorials/launchpad.html

Access to the Course Webpage

- ▶ If you enroll to the course:
 - ▶ Log in at <https://canvas.ust.hk/courses/68288> using your ITSC ID and password
- ▶ If you are auditing this course, please let me know.
- ▶ If you want to get the score for auditing, please finish all homework, exam and project.

Questions?